REAL ESTATE ANALYSIS FOR PROPERTY PURCHASE IN GEORGIA

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Abstract

Our project aims to investigate real estate trends in Georgia. Precisely:

- Investigate the impact of healthcare availability, employment rates, and crime patterns on real estate growth
- Identify potential correlations between these factors and regional growth dynamics
- Showcase findings through a website including dynamic visualizations and interactive dashboard

Hypothesis

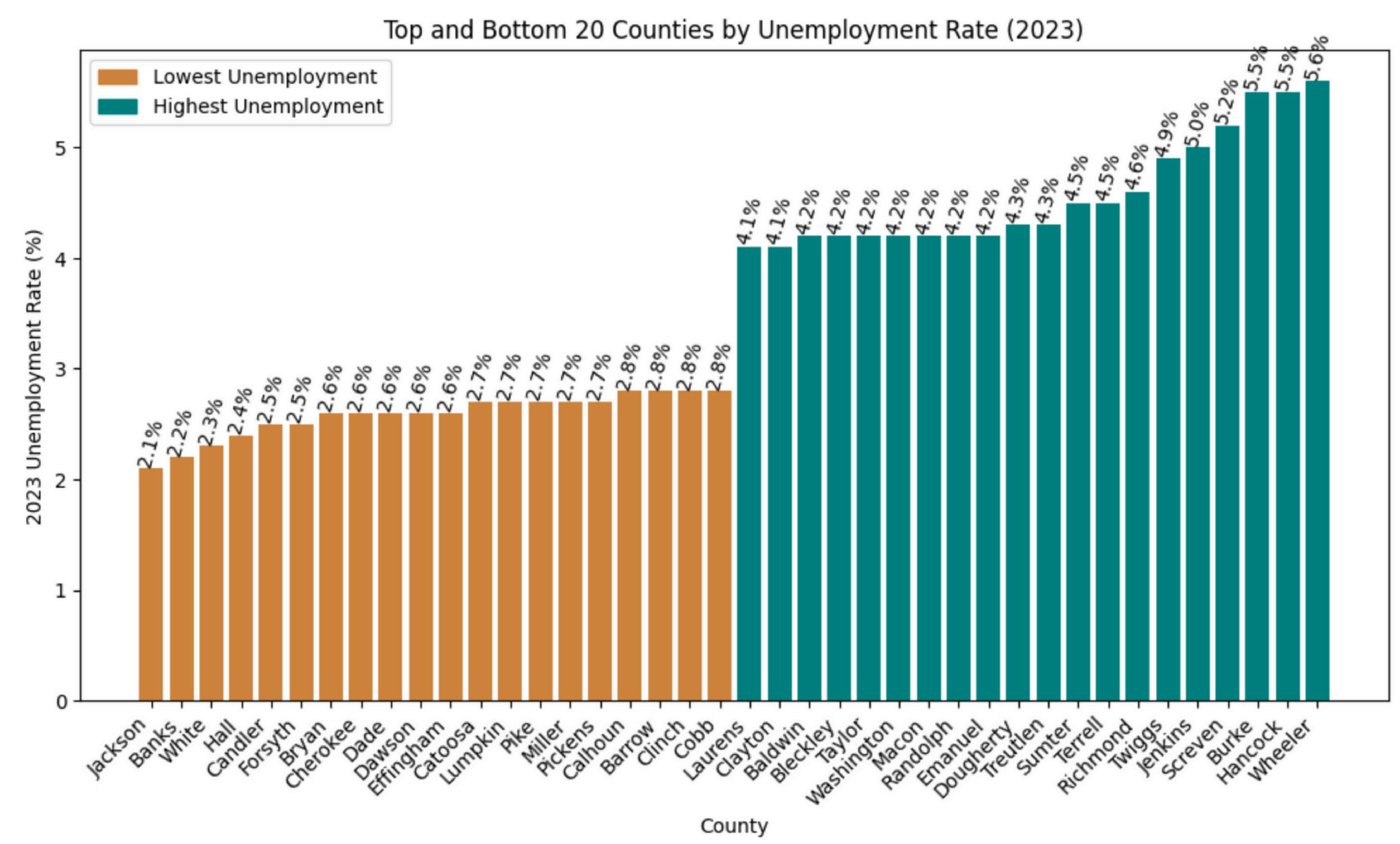
- More hospitals → Higher home prices
- More jobs/businesses → Higher home prices
- Higher crime rates → Lower home prices
- Good community factors ↑ → Real estate values ↑
- Bad factors (like crime) ↑ → Real estate values ↓
- Lower Unemployment ↓ → House Price ↑

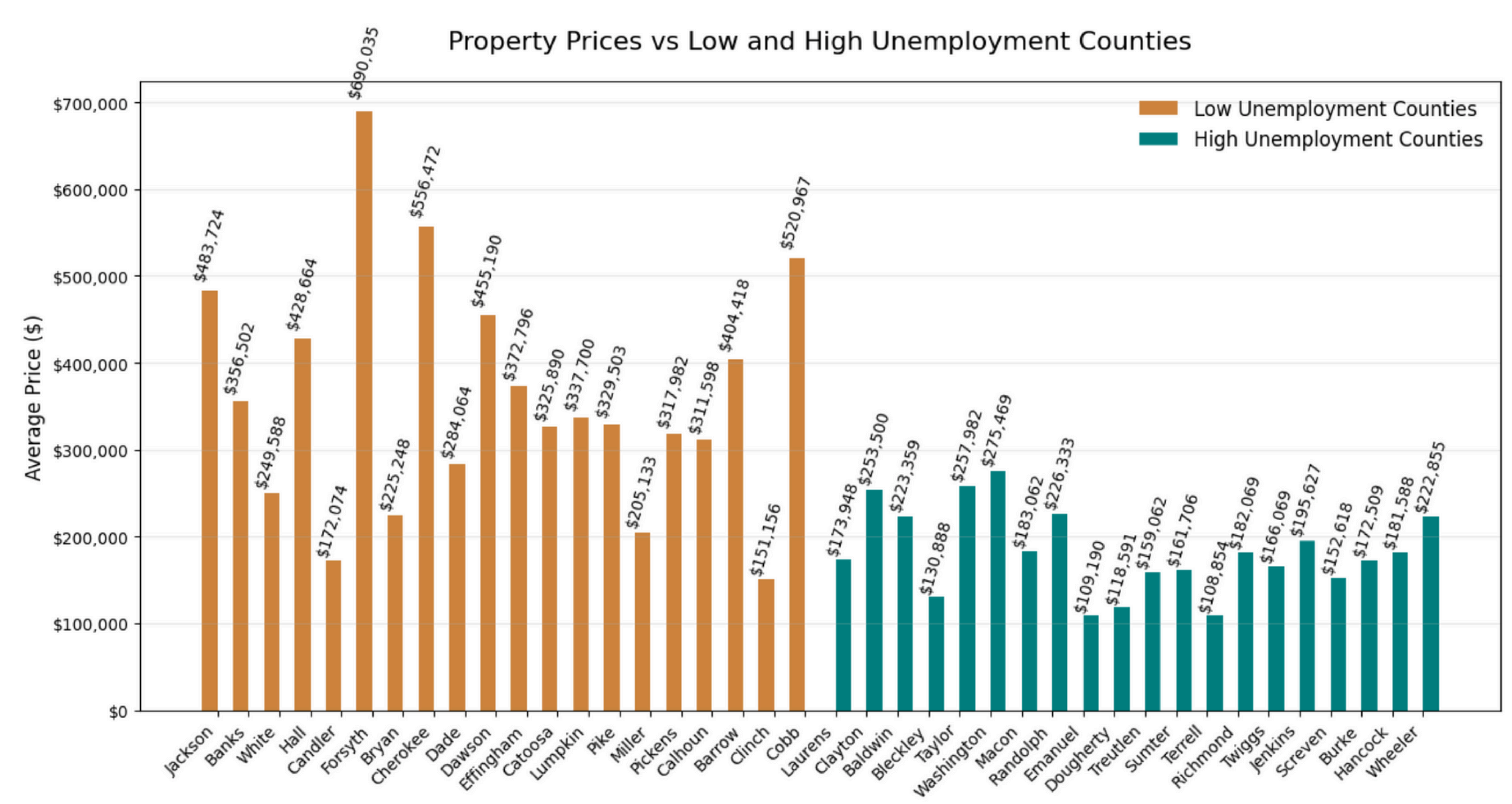
Technologies

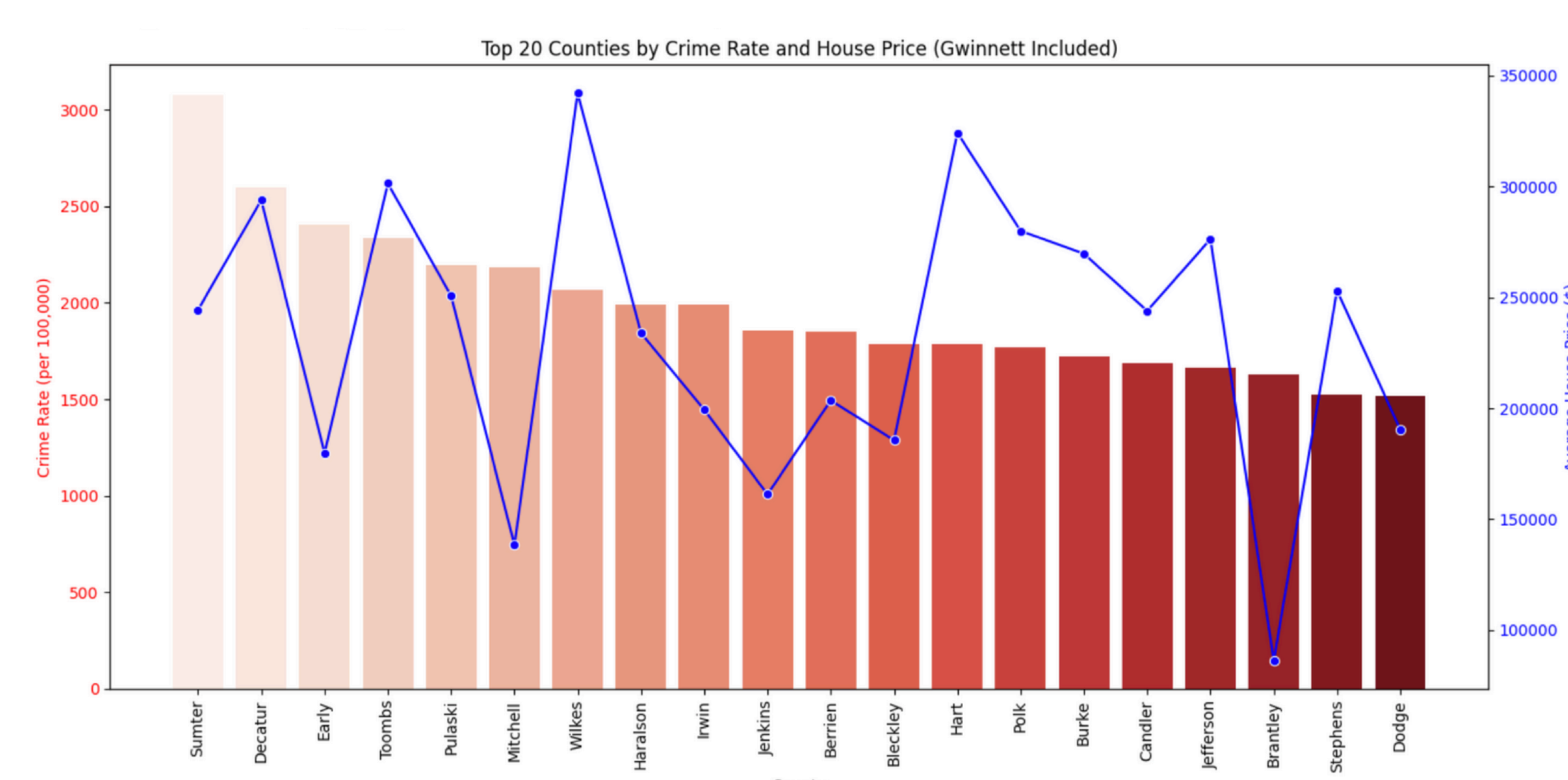
Data Analysis

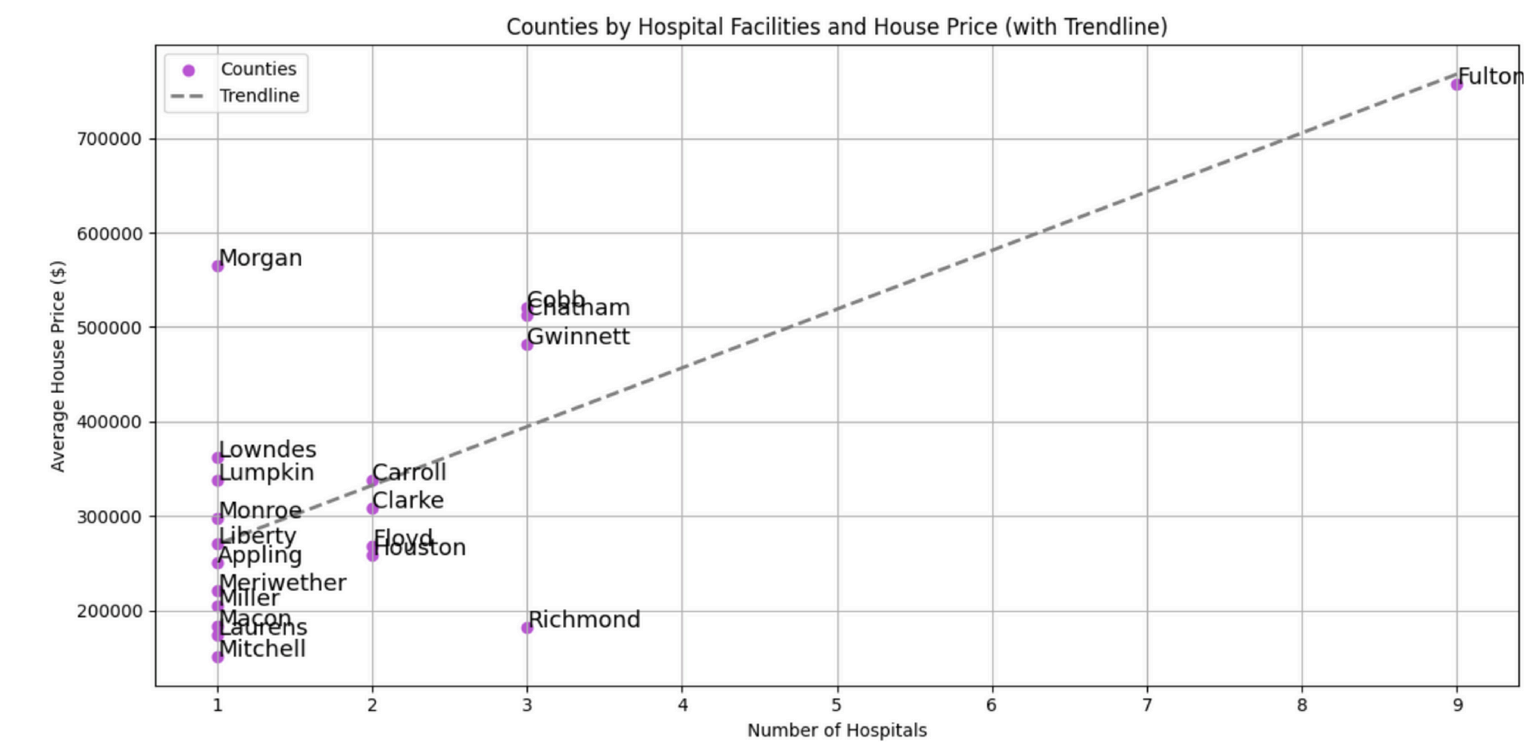
- Gathered real estate data from Kaggle and labor, health, and crime data from Data.gov.
- Converted key variables, such as property prices and crime rates, into numeric formats for accurate analysis.
- Removed duplicates and unnecessary rows
- Standardized county names for consistent merging
- Grouped and summarized data by county for analysis

Results

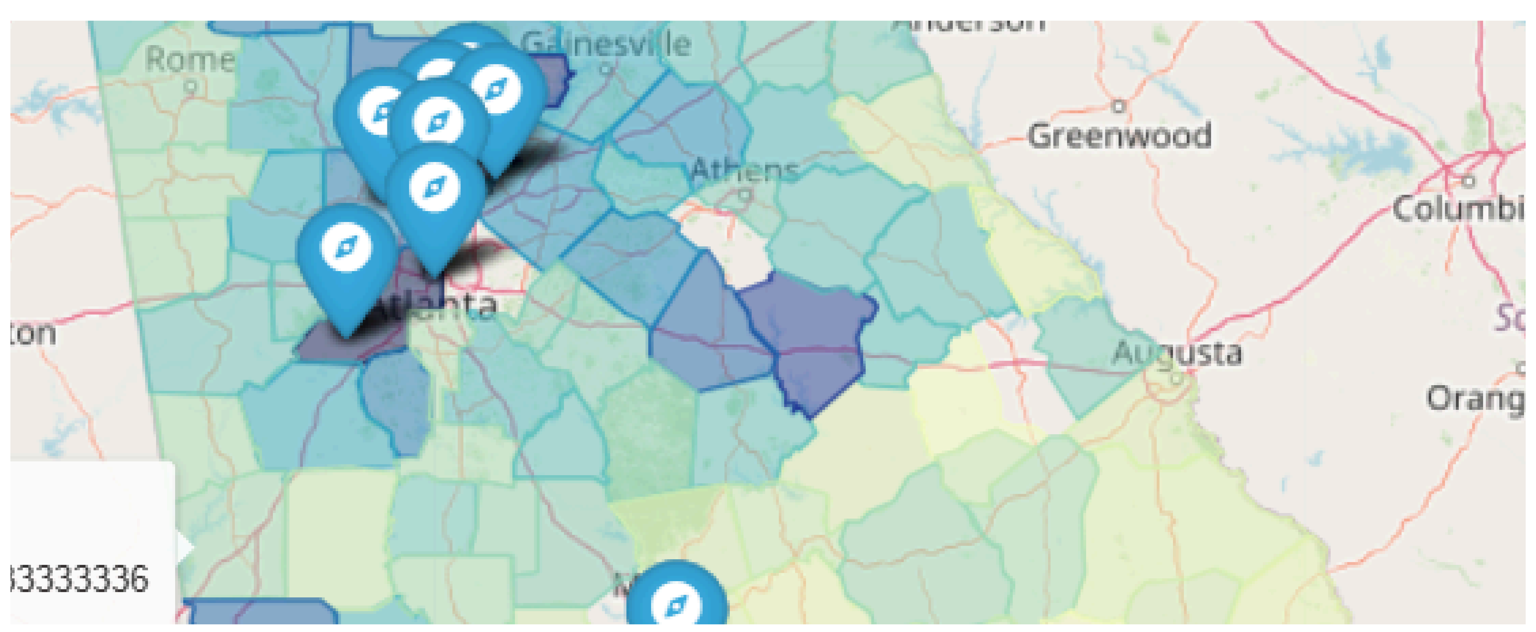








Map of Georgia counties by average property price



Conclusion

What we found:

- low unemployment indicates relatively higher property prices
- close proximity to major cities shows higher property prices
- crime does not seem to be a major factor in determining real estate pricing

Future Work

- Investigate additional links between healthcare, employment, crime, and housing prices across more counties
- Identify stronger correlations using machine learning models like regression and clustering
- Showcase updated findings on the website with visualizations
- Enhance user experience with filters and real-time data integration